Data Analytics Capstone Project

E-Commerce Business Intelligence & Predictive Analytics Platform

Project Overview

You will build an end-to-end data analytics solution for "ShopSmart", a small e-commerce company that sells consumer electronics. The company has been collecting data for 1 year but lacks proper analytics. Your role is to develop a comprehensive analytics platform that provides business insights, basic predictions, and interactive dashboards.

Duration: 3 weeks maximum

Complexity: Intermediate - Integrates all 20 weeks of course content in a streamlined manner

Test Data Sets: **Capstone Test Data sets**

Business Context

ShopSmart has:

- 10,000 registered customers, 5,000 monthly active users
- 5 product categories with 200 SKUs
- Two marketing channels (Email, Paid Ads)
- A new checkout process that needs A/B testing validation Data in CSV files that needs to be organized

Business Problems to Solve:

1. Which products are performing well and which aren't?
2. Can we predict next month's sales?
3. Understanding basic customer purchase patterns

N. Is the new checkout process better than the old one? (A/B test)

U. Build an automated dashboard for weekly monitoring

Project Deliverables

1. Excel Foundation & SQL Database

Deliverable: Combined Excel Analysis + SQL Database

Time: Week 1-2 of the project

Part A: Excel Analysis

- Import sales data (5,000 transactions)
- Clean data: remove duplicates, handle missing values
- Use formulas: SUM, AVERAGE, IF, COUNT, VLOOKUP
- Create ONE pivot table showing monthly revenue by category
- Build a simple dashboard with:
  - Total revenue card
  - Monthly trend line chart
  - Top 5 products bar chart

Part B: SQL Database

- Create 4 simple tables in MySQL/BigQuery:
  - customers (customer_id, name, email, join_date, region)
  - products (product_id, name, category, price)
  - orders (order_id, customer_id, order_date, total)
  - order_items (order_item_id, order_id, product_id, quantity)

- Write 10 essential SQL queries:
  - SELECT with WHERE and ORDER BY
  - Aggregations: SUM, AVG, COUNT by category
  - GROUP BY for monthly revenue
  - INNER JOIN to combine orders with products
  - One subquery to find top customers
  - One CASE statement for customer segmentation
  - One window function (ROW_NUMBER for ranking)

Integration: Establishes your data foundation quickly.

2. Statistical Analysis

Deliverable: Statistical Report (3-5 pages PDF)

Time: Week 3 of the project

Tasks:

- Calculate basic statistics:
  - Mean, median, standard deviation for order values
  - Simple distribution analysis

- Conduct 2 hypothesis tests:
  - **Test 1**: Do email customers spend more than non-email customers? (t-test)
  - **Test 2**: A/B test - does new checkout have better conversion? (proportion test)
  - Document p-values and conclusions
- Simple correlation & regression:
  - Correlation between price and sales volume
  - Linear regression: predict monthly revenue based on number of orders
  - Interpret R-squared

Integration: Validates business decisions with statistical confidence.

3. Python Analysis & Visualization

Deliverable: Jupyter Notebook with Analysis

Time: Week 4-5 of project

Tasks:

Part A: Python Basics & Pandas

- Write 3-4 custom functions (e.g., calculate_retention_rate)
- Load data using Pandas
- Clean data: handle nulls, remove outliers
- Create features: days_since_last_purchase, total_spent
- Group and aggregate data by customer segment

Part B: Visualizations

- Create 6 key charts using Matplotlib/Seaborn:
- Revenue trend line chart
- Category performance bar chart
- Price vs. sales scatter plot
- Correlation heatmap
- Box plot for order values by segment
- Simple cohort retention heatmap

Part C: Simple EDA

- Identify top/bottom 10 products
- Find basic patterns (peak purchase hours, popular categories)
- Detect simple outliers

Part D: One Machine Learning Model

- Build ONE model: Customer churn prediction (will they buy again in 60 days?)
  - Features: recency, frequency, monetary value (RFM)
  - Use Logistic Regression or Random Forest
  - Train/test split: 80/20
  - Evaluate: Accuracy, Precision, Recall
  - Show confusion matrix

Integration: Python automates complex analysis and enables predictions.

4. AI-Assisted Analysis

Deliverable: AI Tools Documentation + AutoML Results

Time: Week 5 of project (parallel with Python)

Tasks:

- Use ChatGPT/Copilot to:
  - Generate 2-3 complex SQL queries
  - Create a Python data cleaning script
  - Generate executive summary text from your analysis

- Use BigQuery ML or Google AutoML to:
  - Build a simple sales forecasting model (predict next month revenue)
  - Compare accuracy with your manual Python model
  - Document: training time, ease of use, accuracy

- Write a 1-page document on:
  - Benefits of AI tools you used
  - Limitations you encountered
  - Basic ethical considerations (data privacy, bias)

Integration: Shows modern analytics leverage AI for efficiency.

5. Interactive Dashboard

Deliverable: Tableau/Power BI Dashboard

Time: Week 6-7 of the project

Tasks:

Build ONE comprehensive dashboard with 3 pages:

Page 1: Business Overview

- Key metrics: Total Revenue, Total Orders, Average Order Value
- Monthly revenue trend
- Revenue by category (bar chart)
- Top 10 products table

Page 2: Product Analytics

- Calculate and display:
  - Monthly Active Users (MAU)
  - Basic Retention Rate (% customers who return)
  - Average Order Value by segment

- Simple funnel: Product View → Add to Cart → Purchase
  - Show conversion % at each step

Page 3: A/B Test Results

- Display checkout process test:
  - Control vs. Variant conversion rates
  - Statistical significance indicator
  - Revenue per user by variant
  - Recommendation box

Interactive Features:

- Add date filter
- Add category filter
- One drill-down feature (click category to see products)
- Tooltips on charts

Integration: Makes insights accessible and interactive.

6. Simple Data Pipeline

Deliverable: Basic dbt Project + Documentation

Time: Week 8 of the project

Tasks:

Simplified Modern Data Stack:

- Load CSV files into BigQuery (or keep in MySQL)
- Create a mini DBT project with just 3 models:

Model 1: stg_orders.sql (Staging)

- Clean order data
- Standardize dates

Model 2: int_customer_metrics.sql (Intermediate)

- Calculate per-customer metrics (total orders, total spend, last order date)

Model 3: mart_monthly_revenue.sql (Mart)

- Create a business-ready monthly revenue summary.
- Include this month vs. last month comparison

- Write 3 simple dbt tests:
  - Unique test on customer_id
  - Not-null test on order_date
  - Check that revenue > 0

- Connect your final dashboard to the dbt mart table

- Create simple documentation:

  - Data flow diagram (source → staging → mart)
  - Brief description of each model
  - How to run dbt (commands)

Integration: Demonstrates understanding of modern data workflow.

Technical Requirements

Tools & Technologies

- **Excel**: Microsoft Excel 2016 or later
- **SQL**: MySQL 8.0+ or Google BigQuery
- **Python**: Python 3.8+ with Jupyter Notebook
- **Python Libraries**: pandas, numpy, matplotlib, seaborn, scikit-learn
- **BI Tools**: Tableau Desktop/Public or Power BI Desktop
- **Cloud Data Warehouse**: Google BigQuery (preferred) or Snowflake
- **dbt**: dbt Core or dbt Cloud
- **AI Tools**: ChatGPT, GitHub Copilot, or Google Bard
- **AutoML**: BigQuery ML or Google AutoML

Data Sources Provided

You will be provided with (or can generate sample data):

1. transactions.csv - 5,000 transaction records (12 months)
2. customers.csv - 10,000 customer records
3. products.csv - 200 product records

N. ab_test_data.csv - A/B test assignment and conversion data (2,000 users)

Note: Sample data will be provided, or you can generate synthetic data using Python/Excel for this project.

Deliverables Checklist

Final Submission Must Include:

1. **Excel Workbook + SQL Scripts** (Combined Deliverable 1)

    - Excel file with data, pivot table, and simple dashboard
    - schema.sql - Database creation script
    - queries.sql - 10 essential queries with comments

2. **Statistical Analysis Report** (Deliverable 2)

    - 3-5 page PDF with hypothesis tests and regression
    - Clear interpretation of results

3. **Jupyter Notebook** (Deliverable 3)

- analysis.ipynb with Python analysis, visualizations, and ML model
- All cells executed with visible outputs

4. **AI Tools Documentation** (Deliverable 4)

- 1-2 page document showing AI tool usage
- AutoML results comparison

5. **Interactive Dashboard** (Deliverable 5)

- .twbx (Tableau) or .pbix (Power BI) file
- 3 dashboard pages as specified

6. **Mini dbt Project** (Deliverable 6)

- dbt folder with 3 models
- Simple documentation (1-2 pages)

7. **Final Presentation**

- 10-15 slide deck with key insights
- Brief video walkthrough (3-5 minutes) - OPTIONAL

8. **README.md**

- Project overview and setup instructions
- Key findings summary

Grading Rubric

Component	Weight	Criteria
Excel + SQL Analysis	20%	Data cleaning, formulas, pivot tables, SQL queries, and database design
Statistical Analysis	15%	Correct hypothesis tests, regression analysis, and clear interpretations
Python Analysis	20%	Pandas usage, visualizations, EDA, ML model performance
AI Integration	10%	Effective use of AI tools, AutoML results, and documentation
Dashboard	20%	Visual design, KPIs, A/B test analysis, interactivity
dbt Pipeline	10%	Model structure, data quality, basic documentation
Overall Quality	5%	Professional presentation, README, clarity

Total: 100%

Timeline & Milestones (3 Weeks)

Week 1-2: Foundation

- **Tasks**: Excel analysis + SQL database + basic queries
- **Milestone**: Excel dashboard complete, SQL database operational with 10 queries

Week 3: Statistical Analysis

- **Tasks**: Hypothesis tests, correlation, regression
- **Milestone**: Statistical report completed (3-5 pages)

Week 4-5: Python Analysis

- **Tasks**: Pandas data manipulation, visualizations, EDA, ML model
- **Milestone**: Jupyter notebook with complete analysis and churn prediction model

Week 5: AI Integration (Parallel)

- **Tasks**: Use AI tools for code generation, AutoML forecasting
- **Milestone**: AI documentation + AutoML model results

Week 6-7: Dashboard Development

- **Tasks**: Build 3-page Tableau/Power BI dashboard
- **Milestone**: Interactive dashboard with KPIs, funnel, and A/B test results

Week 8: dbt Pipeline + Final Polish

- **Tasks**: Create 3 dbt models, connect to dashboard, finalize all deliverables
- **Milestone**: Complete project ready for submission

Success Criteria

Your capstone project will be considered successful if it:

1. **Demonstrates technical proficiency** across all 20 weeks of course content
2. **Provides actionable insights** that ShopSmart can use for business decisions
3. **Shows integration** between different tools and techniques
4. **Includes documentation** that allows another analyst to understand and replicate your work
5. **Presents professionally** with clear visualizations and executive communication
6. **Functions end-to-end** as an automated, scalable analytics platform

Key Business Questions to Answer

Throughout your project, ensure you answer these questions:

1. **Product Performance**: Which products/categories are top performers? Which are underperforming?
2. **Customer Behavior**: What are the basic purchase patterns? (RFM analysis)
3. **Sales Prediction**: Can we predict next month's revenue?
4. **Churn Risk**: Which customers are at risk of not returning?
5. **A/B Test**: Should we keep the new checkout process?

6. **Monthly Trends**: What are the revenue trends over the past year?

Additional Challenges (Optional - For Extra Credit)

If you finish early and want to enhance your project:

1. **Add more visualizations**: Create additional charts in your dashboard
2. **Second ML model**: Build a product recommendation system
3. **More dbt models**: Add 2-3 additional transformation models
4. **Automated email reports**: Use Python to generate weekly summary emails
5. **Geographic analysis**: Add simple maps if customer location data available

Good luck with your capstone project! This is your opportunity to showcase everything you've learned and create a portfolio piece that demonstrates your data analytics expertise.